Johnny Adrián García Betancourt

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Research Profile

Mechatronics engineer and M.Sc. researcher studying the geometric and physical structures of nonlinear dynamical systems. My work draws on differential geometry, geometric mechanics, and port-Hamiltonian systems for modelling and control. I am interested in physics-informed, structure-preserving learning and in the mathematical and dynamical structure of learning itself, with robotics as an application domain.

Education

M.Sc. Student in Electrical Engineering – Control Systems Specialization 2025–Present
King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia GPA: 3.938/4.00
Expected Completion: January 2027.
Thesis: *Bridging Physical Safety and Active Tolerance in Robotic Manipulators: A Hierarchical Energy-Aware Architecture.*
Selected Coursework: Geometric Control (**A+**), Classical Mechanics (**A+**), Advanced Methods for Theoretical Physics (**A+**), Linear Control Systems (**A+**), Optimal Control (**A+**), Robotics and Control (**A**), Image Processing (**A+**).

Graduate Coursework in Automation and Control (Master's Program) 2024
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador Average: 9.68/10
Selected Coursework: Identification and Control of Dynamical Systems (A+), Machine Learning-Based Control (A+), Industrial Robotics and Computer Vision (A+), Industrial Instrumentation and Control (A+), Embedded Control Systems (A+).

B.Sc. in Mechatronics Engineering (Engineer in Mechatronics) 2017–2022
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador Thesis Grade: 9.20/10
Thesis: *Mechanical and Electrical Design, and Automation of an Industrial Liquid Filling Machine.*

Research Experience

Spectral-Hamiltonian Algebraic Power Evaluation (SHAPE Framework)
KFUPM | First-author manuscript [1] | Preprint | Code
- Formulated algebraic port-power residuals from port-Hamiltonian relations and differential flatness, without a dedicated residual observer; generated simulation data and co-designed/implemented multi-joint PyTorch CNNs.
- On a simulated 6-DOF Kinova Gen3, achieved 84.69% mean accuracy across five training seeds and 3,000 held-out runs with CWT inputs; characterized weak-fault and velocity-noise limitations.

Hierarchical Energy-Aware Architecture for Robotic Manipulators
Ongoing M.Sc. thesis | KFUPM | Advisors: B. Brahmi and R. Rashad
- Developing a 7-DOF Panda architecture connecting geometric replanning on $(S^1)^7$, partitioned tanks with energy freezing and transfers, and actuator-capacity checks under declared loss of effectiveness.
- Analyzed finite valve closure for a liquid-energy-driven clock with consistent feedforward; evaluated one prepared nominal recovery episode with a 120-s terminal hold in simulation.

Flatness-Based Predictive Control under Partial Actuator Degradation
KFUPM | First-author manuscript [2] | Code
- Developed a sampled flatness-based MPC realization with a local affine held-torque constraint map, a disturbance observer, and path-preserving timing supervision.
- Benchmarked against LTV MPC in 232 Kinova simulations; reduced mean paired command TV during reference motion by 6.0%/6.7% on two paths, with 30 paired uncertainty realizations per path.

Geometric Modelling & State Estimation of an Underwater Glider
Coursework & collaborative research | KFUPM | Code
- Led collaborative $\mathrm{SE}(3)$-based glider modelling with articulated internal masses, syringe-based variable ballast, and configuration-dependent inertia; simulated gliding and helical motion in MATLAB.
- Accounted for open-system energy exchange; in a separate Simulink coursework model, implemented Kalman filtering for depth and vertical-velocity estimation.

Geometric Dynamics Simulators – Franka Panda & Kinova Gen3
Research software | Panda repository | Kinova repository
- Developed MATLAB models of the 7-DOF Panda and 6-DOF Kinova with screw-theoretic $\mathrm{SE}(3)$ kinematics, spatial RNEA dynamics, and torus-based configurations.
- Implemented Lie-group integration and verified dynamics against Lagrangian/COM-Jacobian references, with kinematic, power-balance, and manifold-consistency checks.

Research Manuscripts & Working Papers

[1] J. Garcia, N. Salguero, N. Iqbal, B. Brahmi, and R. Rashad, "Energy-Based Fault Detection and Isolation Framework for Flat Port-Hamiltonian Mechanical Systems," *arXiv preprint* arXiv:2610.33679, 2026. Submitted to *IEEE Transactions on Control Systems Technology*; decision pending.

[2] J. Garcia, N. Salguero, B. Brahmi, and N. Iqbal, "Flatness-Based Predictive Control of Robotic Manipulators under Partial Actuator Degradation," manuscript, 2026. Submitted to SSD 2027; decision pending.

[3] U. I. Castilla Batun and J. A. Garcia Betancourt, "Techno-Economic Feasibility of Wind-Powered Hydrogen Production in Saudi Arabia: A Multi-Regional Assessment," research manuscript, 2026. Being revised for resubmission.

Technical Expertise

Mathematical Foundations	Linear and tensor algebra, differential geometry, ODEs/PDEs, calculus of variations, probability and statistics, and mathematical optimization.
Physical Foundations	Classical and geometric mechanics, Lagrangian/Hamiltonian formulations, symplectic and Poisson structures, conservation laws and symmetries, port-Hamiltonian systems, and passivity.
Control & Robotics	Nonlinear and geometric control, stability analysis, differential flatness, optimal control and MPC, robotic dynamics, trajectory optimization, and fault-tolerant control. Screw-theoretic kinematics and the recursive Newton–Euler algorithm (RNEA) for inverse dynamics in Kinova/Franka models.
Machine Learning & AI	Physics-guided machine learning and feature engineering, deep neural networks (1D/2D CNNs), multi-joint feature fusion, model benchmarking, robustness evaluation, and uncertainty quantification.
State Estimation & Signal Processing	Bayesian and invariant state estimation, Lie-group methods, Fourier and time-frequency analysis, digital filtering, image processing, and feature extraction.
Programming & Tools	Python, C++, PyTorch, MATLAB/Simulink, Simscape Multibody, Git, LaTeX, SolidWorks (CAD), and ANSYS (CAE).
Robotic Platforms	Quanser QArm: vision-based grasping; Kawasaki RS03N at ESPOL: Python-based pick-and-place; Franka Emika: basic hardware motion tests.

Academic and Professional Experience

Teaching Assistant – Electronics and Instrumentation Labs 2025–Present
KFUPM, Saudi Arabia. Led undergraduate labs on circuits, signal conditioning and electronic measurements.

Teaching Assistant – Statistics and Probability 2022
ESPOL, Ecuador. Led problem-solving sessions; authored midterm/final exam problems and graded assignments.

Maintenance and Automation Coordinator 2021–2024
Detecuador S.A., Ecuador. Led industrial modernization, CAD design, automation, instrumentation and condition monitoring.

Assistant Electromechanical Designer 2023–2024
SEOR S.A., Ecuador. Electromechanical CAD, control cabinets and prototypes; sensor integration and PLC programming.

Engineering Projects

Industrial Process Automation & Condition Monitoring Graduate Project | ESPOL / Detecuador
Led an approximately USD 25,000 liquid-processing project: recipe-based mixing, PLC/HMI automation, level monitoring and IO-Link. Built a separate motor test bench for vibration, temperature and speed; explored MATLAB predictive-maintenance workflows.

Industrial Liquid-Filling Machine B.Sc. Thesis | ESPOL | 2022
Designed and modelled a filling machine for viscous products: pneumatic/fluid calculations for dosing and cycle timing, electrical sizing, PLC/HMI control, and CAD/CAE and finite-element analysis of the hopper and structure.

Honors and Awards

Graduate Full Scholarship, KFUPM 2025–Present
Best Mechatronics Thesis Award (1st Place), ESPOL 2022
Best University Thesis Award (2nd Place Institution-Wide), ESPOL 2022
Academic Excellence Recognition (Top Score), National University Admission Exam (SENESCYT) 2017

Languages

Spanish – Native **English** – Advanced / Academic working proficiency

Academic References

Dr. Ramy Rashad – Assistant Professor
CIE, KFUPM • ramy.rashad@kfupm.edu.sa
M.Sc. Co-advisor & Geometric Control Instructor

Dr. Brahim Brahmi – Assistant Professor
EE, KFUPM • brahim.brahmi@kfupm.edu.sa
M.Sc. Thesis Advisor & Optimal Control Instructor